

CivicDividendOS: A Factor-Origin Fiscal Ledger and Public Automation Dividend Framework for Human–AI–Robot Economies

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Abstract—The rapid diffusion of artificial-intelligence agents, autonomous software, and physical robots is beginning to weaken the historical correspondence between productive work, human wages, payroll taxation, and household income. Existing automation-tax research largely asks whether robots should be taxed and at what rate, while recent AI public-finance research focuses on capital taxation, corporate profits, consumption taxes, transfers, or sovereign wealth funds. These approaches are valuable but leave an unresolved accounting and distribution problem: when an urban service, firm, or digital platform is jointly operated by humans, AI agents, robots, data, and conventional capital, public authorities lack a principled mechanism for determining which factor created economic value, who should legally bear the fiscal obligation, and how automation-generated surplus should be distributed between private owners, workers, citizens, and the state.

This paper proposes *CivicDividendOS*, a novel factor-origin fiscal and distribution framework for economies in which human labour coexists with autonomous digital and embodied productive systems. The framework introduces an *Autonomous Economic Activity Passport* for every significant AI-agent or robotic deployment. The passport does not grant legal personality to machines; instead, it creates an auditable economic identity linking machine activity to its human or corporate owner, operator, jurisdiction, energy use, task class, and generated output. A *Factor-Origin Ledger* then attributes value added across human labour, AI-agent services, robotic services, data contributions, and conventional capital using task-level causal contribution estimates rather than simply imputing a fictional wage to a robot.

Based on this ledger, *CivicDividendOS* computes a dynamic *Automation Contribution Base* and applies a *Social Automation Contribution* to the legally responsible owner or beneficiary. The rate is endogenous: it rises when autonomous production substitutes for taxable labour, concentrates economic rents, or creates measurable social externalities, and falls through augmentation, reskilling, human-capability, small-firm, and productivity credits when automation complements workers or produces broad social benefit. Collected revenue and selected equity contributions are divided through a *Public Automation Wealth Fund* into four channels: a universal Civic Automation Dividend, targeted worker-transition insurance, a Human Capability Endowment for education and future expertise, and financing for public services. In parallel, a *Human Earned-Income Shield* reduces payroll or labour-income burdens as the taxable labour share contracts.

The framework therefore changes the fiscal unit of analysis from “taxing robots” to *taxing the economically attributable automation surplus at the owner/beneficiary level while socializing part of its long-run capital return*. A policy digital twin continuously stress-tests tax incidence, innovation effects, employment, inequality, revenue stability, and cross-border leakage before rates are adjusted. This provides a computational foundation

for a tax-and-dividend system that can preserve public revenue and household purchasing power without treating AI agents or robots as fictional legal persons.

Index Terms—AI taxation, robot tax, automation, fiscal policy, public finance, multi-agent systems, income distribution, universal basic capital, sovereign wealth fund, future of work, agentic economy, social dividend.

I. INTRODUCTION

Modern tax systems were largely designed around economies in which human work generates wages, employers remit payroll contributions, households pay income and consumption taxes, and owners receive taxable profits from capital. Automation has always complicated this structure, but agentic AI and increasingly autonomous robots introduce a qualitatively broader challenge. Software agents can now execute cognitive, administrative, creative, analytical, and commercial tasks, while robots increasingly perform physical activities previously requiring human workers.

The economic output of a future firm or city service may therefore arise from a joint production process involving humans, AI agents, robots, training data, cloud infrastructure, intellectual property, and conventional capital. If labour’s share of production falls, governments may face a structural mismatch: the tax base remains strongly linked to wages and payroll while economic value increasingly accrues to owners of automated productive capital.

The policy question is not simply whether a robot should “pay tax.” A robot or AI agent does not currently possess tax residence, citizenship, consumption needs, pension rights, or legal responsibility analogous to a human worker. Current legal analysis generally treats AI and robots as assets, tools, services, or components of businesses rather than independent taxable persons [17]. The more important problem is how to attribute economic value produced through autonomous systems and how to distribute a reasonable share of the resulting surplus without destroying socially valuable innovation.

This paper proposes *CivicDividendOS*, a computational framework that combines factor-origin accounting, adaptive automation contributions, public capital ownership, citizen dividends, worker transition support, and labour-tax relief.

II. LITERATURE REVIEW

A. Automation, Tasks, Wages, and Income Distribution

A central insight in the economics of automation is that technology changes the allocation of tasks between labour and capital. Acemoglu and Restrepo show that automation produces a displacement effect by allowing capital to perform tasks previously undertaken by labour, while the creation of new tasks can reinstate labour demand [1]. Their task-based perspective provides an important foundation for analysing fiscal policy because automation affects not only aggregate productivity but also who receives the associated income.

Empirical research further shows that the labour-market effects of robot adoption are uneven. Acemoglu and Restrepo document adverse employment and wage effects in U.S. local labour markets exposed to industrial robots [2]. By contrast, Dauth et al. find that manufacturing job displacement in Germany was offset by service-sector employment, although the burden fell disproportionately on younger workers entering the labour market [3]. These findings suggest that automation can generate both gains and adjustment costs, and that distributional consequences differ across time, workers, and institutional environments.

Recent AI research broadens these concerns because AI can affect non-routine cognitive tasks in addition to routine physical or clerical work. The IMF has emphasized that AI may increase productivity while also widening income and wealth inequality if gains disproportionately accrue to high-income workers and capital owners [11]. OECD evidence similarly indicates heterogeneous effects across occupations and skill groups [14].

B. Robot Taxes and Optimal Taxation

The idea of taxing robots has generated substantial economic and legal debate. Guerreiro, Rebelo, and Teles develop a quantitative model in which falling automation costs can sharply increase inequality. They find that robot taxation can be optimal during the transition while workers exposed to automation remain in the labour force, but that the optimal tax can disappear after adjustment [5].

Thuemmel studies the joint taxation of robots, other capital, and labour income. The analysis shows that the optimal treatment of robots depends on their cost and general-equilibrium effects: subsidies may be desirable when robots are expensive, while a positive robot tax can emerge as robots become cheaper and inequality increases [6]. This demonstrates that the optimal policy is dynamic rather than a permanent fixed levy.

Acemoglu et al. analyse optimal policy in a task framework and show that reducing labour taxes while combining capital-tax reductions with targeted automation taxes can improve employment outcomes relative to uniform capital taxation [7]. Related work on inefficient automation finds that firms may automate too rapidly because they do not internalize transition costs borne by workers, creating an efficiency rationale for slowing automation even without an explicit redistributive objective [8].

Merola reviews taxation as a mechanism for sharing gains from automation and AI, discussing robot taxes, capital taxation, and redistribution toward workers [10]. Zhang similarly analyses automation, wage inequality, and robot taxation [4].

The literature therefore establishes that some form of differentiated taxation of automation may be justified under particular conditions. However, a persistent implementation challenge remains: what exactly constitutes a robot, how should its contribution be measured, and how can the tax avoid discouraging beneficial capital deepening?

C. Automation and the Tax Base

Automation can alter not only employment but also the structure of government revenue. Hötte, Theodorakopoulos, and Koutroumpis examine nineteen European countries and find that the relationship between automation and tax receipts varies across technologies and phases of diffusion [9]. Their results caution against the simplistic assumption that automation necessarily causes a permanent decline in total revenue, but they also demonstrate that automation can change the balance between labour, capital, and consumption tax bases.

The IMF argues that fiscal policy will need to adapt if AI shifts income toward capital and away from labour, including stronger social protection and careful consideration of capital-income taxation [12]. The IMF's 2026 work on the global economic implications of AI similarly emphasizes protecting tax capacity and fairness if labour income becomes less central [13].

Korinek and co-authors develop a broader public-finance framework for advanced AI economies, analyzing optimal taxation under scenarios in which capital can become increasingly dominant [15]. This line of work is especially important because it moves beyond a narrow robot levy toward the redesign of the fiscal system itself.

D. Taxing AI Rather than Only Robots

The expansion of generative and agentic AI has renewed interest in taxation instruments that are not tied to physical robots. Faivre and Cen survey the viability of AI taxation and distinguish multiple objectives, including correcting externalities, redistributing uneven gains and costs, and financing regulatory capacity [16]. They discuss corporate income and rent-based taxation, consumption taxes on AI services, and excise-style instruments tied to particular AI activities.

This work highlights an important design principle: there is unlikely to be one universal "AI tax." Different harms and policy objectives require different tax bases. Measurement, tax incidence, international leakage, and innovation effects are major constraints.

Legal scholarship similarly identifies substantial difficulties in treating robots or AI systems as independent taxpayers. Current income-tax systems generally attribute income to owners, firms, or service providers rather than to the autonomous system itself [17]. This creates a strong case for separating *machine-level economic measurement* from *legal tax liability*.

E. Universal Basic Income, Social Protection, and Transfers

One proposed response to automation is redistribution through universal basic income (UBI) or targeted transfers. Merola notes that revenues from automation taxes could be used for universal or worker-focused transfers [10]. The ILO has examined UBI proposals in relation to social protection standards and has highlighted financing, adequacy, labour-market, and institutional questions [18].

Transfers can stabilize household income, but they do not necessarily change the underlying ownership of productive capital. If AI-driven production becomes capital intensive, a purely transfer-based model may require governments to continually tax a concentrated ownership base.

F. Universal Basic Capital and Public Wealth Funds

An alternative is broader ownership of productive assets. Universal Basic Capital (UBC) proposes granting citizens financial assets or capital rather than only periodic income transfers [19]. Progressive sovereign wealth fund proposals similarly argue that public ownership of capital can generate a more egalitarian distribution of investment income [20].

The Alaska Permanent Fund provides a long-running real-world example of converting natural-resource rents into a permanent investment fund whose earnings support resident dividends [21]. Although AI is not a natural resource, the institutional analogy is relevant: temporary or concentrated economic rents can be partially transformed into broadly owned financial capital.

Recent public-finance discussions increasingly consider sovereign wealth funds or citizen-share funds as complements to AI-era tax reform. Brookings argues that equity-based mechanisms can capture returns that scale with AI's economic importance [22], [23].

G. Data Dividends

Another literature considers whether individuals should receive dividends from data used to create AI value. Vincent et al. map design choices for data dividends and show that apparently reasonable allocation rules can generate concentrated or demographically unequal outcomes [24]. Bax investigates Shapley-value and Owen-value approaches for estimating the contribution of individual data samples [25].

This literature is useful for value attribution, but data dividends address only one input into AI production. They do not provide a comprehensive fiscal system for joint human–AI–robot production.

H. Adaptive and Agent-Based Fiscal Policy

AI itself is also being investigated as an instrument for fiscal design. TaxAgent integrates large language models with agent-based economic simulation to adapt tax policies under heterogeneous household behaviour [26]. Other research explores AI-supported detection of tax loopholes and policy inconsistencies [27]. These developments suggest that future fiscal systems may be increasingly adaptive and simulation-driven.

However, these systems typically optimize conventional tax structures. They do not establish an accounting layer that identifies the factor origin of value in mixed human–agent–robot production.

III. SYNTHESIS OF THE LITERATURE

The literature provides five strong foundations:

- 1) automation changes the task allocation between capital and labour;
- 2) automation can affect wages, employment, inequality, and tax bases;
- 3) targeted taxation of automation may be optimal under some conditions;
- 4) social transfers and public capital ownership can distribute automation gains; and
- 5) adaptive computational methods can support tax-policy design.

Nevertheless, these literatures are generally connected only at a macroeconomic or conceptual level. A practical fiscal architecture for an agentic economy requires a missing intermediate layer: *auditable attribution of economic value to the productive factors that actually performed a task.*

IV. RESEARCH GAP

Three unresolved gaps motivate CivicDividendOS.

A. The Attribution Gap

Existing robot-tax proposals frequently rely on robot counts, robot investment, imputed wages, capital expenditure, or firm-level profits. These measures become increasingly problematic in an agentic economy. One AI agent may execute the work of many humans at negligible marginal software cost, while another AI system may simply augment employees. A robot may perform a hazardous task that should be encouraged rather than taxed. A cloud-based agent may serve multiple jurisdictions simultaneously.

Therefore, neither “one robot = one taxable worker” nor a simple tax on automation investment provides a reliable measure of the economic value created by autonomous systems.

B. The Liability Gap

Giving AI agents or robots fictional salaries and asking them to pay income tax creates legal and administrative difficulties. Machines do not independently own their output under current systems. The economically relevant beneficiary is generally an owner, firm, platform, or operator.

A new framework must therefore identify machine-originated economic activity without making the machine itself the legal taxpayer.

C. The Distribution Gap

Current proposals usually choose one redistribution mechanism: a robot tax, higher capital taxation, UBI, retraining, or public ownership. Yet an AI economy can require several simultaneously. Displaced workers have immediate transition needs; future workers need education; citizens may need a new

non-wage source of income; governments need stable revenue; and innovation still requires private returns.

The literature does not yet provide an integrated, task-origin fiscal mechanism that determines both *how automation surplus enters the public system* and *how that surplus is divided among current citizens, affected workers, future human capability, and public capital*.

V. PROBLEM STATEMENT

Let economic output Y be jointly produced by human labour H , AI-agent services A , robotic services R , data D , and conventional capital K :

$$Y = F(H, A, R, D, K). \quad (1)$$

Under a traditional fiscal system, a substantial share of public revenue is generated from:

$$Tax^{traditional} = \tau_L W_H + \tau_K \Pi + \tau_C C, \quad (2)$$

where W_H is human wage income, Π represents taxable capital or corporate income, and C consumption.

As autonomous productive factors replace or augment human tasks, the wage share may decline while output remains constant or increases:

$$\frac{W_H}{Y} \downarrow, \quad \frac{Y_A + Y_R}{Y} \uparrow. \quad (3)$$

If ownership of A and R is concentrated, household purchasing power, payroll-financed social protection, and government revenue can weaken even when aggregate productivity rises.

The fundamental research problem is therefore:

How can a tax-and-distribution system identify the economic value created by humans, AI agents, robots, data, and conventional capital; assign legal fiscal responsibility to the appropriate human or corporate beneficiaries; and distribute part of automation-generated surplus in a way that preserves innovation, public revenue, household income, workforce transition, and broad participation in future capital returns?

The problem becomes more difficult when AI agents are cloud-based, operate across borders, change tasks rapidly, and act primarily as complements rather than substitutes for humans.

VI. PROPOSED FRAMEWORK: CIVICDIVIDENDOS

CivicDividendOS contains seven interconnected components.

A. 1. Autonomous Economic Activity Passport

Every economically significant autonomous deployment is assigned an *Autonomous Economic Activity Passport* (AEAP).

The passport is not legal personhood. It is a fiscal measurement record:

$$AEAP_j = \{ID_j, O_j, J_j, T_j, E_j, Q_j, R_j, M_j\}, \quad (4)$$

where: O_j is the legally responsible owner/operator, J_j deployment jurisdiction, T_j task category, E_j energy/resource consumption, Q_j measured output, R_j risk classification, and M_j model/robot class.

The AEAP allows economic activity generated by an AI agent or robot to be audited without pretending that the system itself is a taxpayer.

B. 2. Factor-Origin Ledger

For every production episode p , CivicDividendOS estimates the marginal or causal contribution of five productive factors:

$$\phi_p = \{\phi_H, \phi_A, \phi_R, \phi_D, \phi_K\}, \quad (5)$$

subject to

$$\sum_f \phi_f = 1. \quad (6)$$

The attributable value generated by factor f is:

$$V_{p,f} = \phi_{p,f} V_p. \quad (7)$$

Contribution estimates may be obtained through controlled counterfactuals, production-function estimation, Shapley-style contribution methods, matched historical baselines, or digital-twin simulation.

This is a central departure from traditional robot taxation: the system taxes neither the physical count of robots nor a fictional machine wage. It estimates the economic origin of value.

C. 3. Substitution–Augmentation Classifier

Automation should not be treated equally when it replaces a worker and when it makes a worker more productive.

Define the automation mode:

$$Z_p \in \{\text{Substitution, Augmentation, NewTask, SafetyReplacement}\}. \quad (8)$$

A substitution score can be estimated as:

$$S_p = \Pr(\Delta H_p < 0 \mid A_p, R_p, \mathbf{X}_p), \quad (9)$$

while an augmentation score measures gains to human productivity or wages.

This classification feeds directly into the fiscal rate.

D. 4. Automation Contribution Base

The proposed *Automation Contribution Base* (ACB) is:

$$ACB_i = VA_i^{A,R} - C_{A,R,i}^{allowable}, \quad (10)$$

where $VA_i^{A,R}$ is value added attributable to AI-agent and robotic factors and $C_{A,R,i}^{allowable}$ represents verified operating, compute, energy, depreciation, and acquisition costs according to policy.

The legal taxpayer is:

$$Taxpayer_i = Owner/Operator/Beneficiary(AEAP_i), \quad (11)$$

not the AI agent or robot.

E. 5. Dynamic Social Automation Contribution

The *Social Automation Contribution* (SAC) is:

$$SAC_i = r_i \cdot ACB_i, \quad (12)$$

where the adaptive rate r_i is:

$$r_i = r_0 + \alpha S_i + \beta C_i^{rent} + \gamma E_i^{disp} + \delta X_i^{ext} + \eta R_i^{revenue} - \theta A_i^{aug} - \lambda T_i^{train} - \mu B_i^{broad} - \nu N_i^{newtask}. \quad (13)$$

Here:

- S_i = labour substitution intensity;
- C_i^{rent} = economic-rent/concentration score;
- E_i^{disp} = measurable worker-displacement burden;
- X_i^{ext} = environmental or social externalities;
- $R_i^{revenue}$ = erosion of payroll/social-contribution base;
- A_i^{aug} = human augmentation benefit;
- T_i^{train} = verified worker training and transition investment;
- B_i^{broad} = broad-based ownership/benefit score; and
- $N_i^{newtask}$ = creation of new human tasks.

Thus automation that complements labour can face a low or even zero incremental contribution, while high-rent labour-replacing automation can face a higher rate.

F. 6. Human Earned-Income Shield

A core innovation is to use automation revenue partly to reduce the tax wedge on human work.

Let the labour-income tax relief rate be:

$$\Delta\tau_L(t) = -\kappa \frac{Rev_{SAC}(t)}{TaxBase_L(t)}. \quad (14)$$

As autonomous production contributes more to public revenue, qualifying low- and middle-income human workers receive lower payroll or earned-income tax burdens.

This avoids a system in which society simultaneously taxes humans heavily for working while giving machines a zero marginal social-contribution burden.

G. 7. Public Automation Wealth Fund

A fixed fraction of SAC revenue and selected equity-based contributions enter a professionally managed *Public Automation Wealth Fund* (PAWF):

$$PAWF_{t+1} = PAWF_t(1+r_t^{fund}) + \omega Rev_{SAC,t} + Equity_t - Dist_t. \quad (15)$$

The fund converts a portion of temporary AI and automation rents into long-lived public capital.

Annual distributable returns are divided among four accounts:

$$Dist_t = D_t^{citizen} + D_t^{transition} + D_t^{capability} + D_t^{public}. \quad (16)$$

1) *A. Civic Automation Dividend*: Every eligible resident receives a base dividend from investment returns rather than directly consuming all annual tax receipts:

$$CADiv_t = \frac{\rho D_t^{citizen}}{N_t}. \quad (17)$$

This creates a non-wage household income stream linked to the productivity of the automated economy.

2) *B. Worker Transition Insurance*: Workers experiencing verified displacement receive temporary wage insurance, mobility support, or reskilling finance:

$$WTI_i = g(WageLoss_i, Tenure_i, Exposure_i, Training_i). \quad (18)$$

3) *C. Human Capability Endowment*: Part of the return finances education, apprenticeships, advanced technical skills, public-service expertise, and human capabilities likely to complement future AI.

4) *D. Public-Service Account*: A final portion supports social insurance and public services whose funding previously depended heavily on payroll contributions.

VII. AUTOMATION OWNERSHIP UNITS

CivicDividendOS adds a predistribution mechanism rather than relying exclusively on ex-post tax transfers.

When a qualifying firm receives major public AI subsidies, public compute, exclusive public data access, or government procurement support, a policy may require a small *Automation Ownership Unit* contribution:

$$AOU_i = \xi_i \cdot Equity_i. \quad (19)$$

These units enter the PAWF rather than individual government departments.

This mechanism resembles the logic of a resource fund: the public contributes infrastructure, research, data, procurement demand, education, and legal institutions that help create private automation value, and therefore may hold a modest diversified claim on future returns.

VIII. DISTRIBUTION WATERFALL

The complete income-distribution mechanism can be represented as:

$$Y = Y_H + Y_A + Y_R + Y_D + Y_K. \quad (20)$$

Private primary distribution remains:

$$Y^{private} = W_H + \Pi_{owners} + R_D + R_K. \quad (21)$$

CivicDividendOS then introduces a secondary and ownership-based distribution:

$$Y_H^{final} = W_H - Tax_H + Shield_H + Dividend_H + Transition_H + Services_H, \quad (22)$$

while owners receive:

$$Y_O^{final} = \Pi_O - CIT_O - SAC_O + PrivateReturn_O. \quad (23)$$

The state/public receives:

$$Y_P = CIT + SAC + PAWFReturns, \quad (24)$$

which is split among services, human-capability investment, transition support, and citizen dividends.

The design therefore preserves private profit incentives while creating a parallel public claim on part of the automation surplus.

IX. FISCAL DIGITAL TWIN

Before changing SAC rates or distribution shares, policy is tested in a fiscal digital twin containing heterogeneous households, firms, AI agents, robots, sectors, and jurisdictions.

The policy vector is:

$$\Theta = \{r_0, \alpha, \beta, \gamma, \omega, \rho, \tau_L, \xi, \text{credits}\}. \quad (25)$$

The policy engine seeks:

$$\Theta^* = \arg \max_{\Theta} [Growth + Revenue + Welfare + Employment + Innovation - Inequality - Leakage] \quad (26)$$

subject to fiscal-sustainability and minimum-innovation constraints.

The digital twin must report tax incidence explicitly. A nominal tax on an AI operator may ultimately fall on shareholders, workers, suppliers, or consumers; therefore the policy is not adjusted from statutory rates alone.

X. CROSS-BORDER DEPLOYMENT NEXUS

Cloud AI complicates conventional tax residence because an agent can be trained in one jurisdiction, owned in another, hosted in a third, and economically deployed in a fourth.

CivicDividendOS therefore proposes a *Deployment Nexus*:

$$Nexus_{ij} = w_1 Usage_{ij} + w_2 Revenue_{ij} + w_3 Users_{ij} + w_4 PhysicalOps_{ij}. \quad (27)$$

Automation contribution revenue can then be apportioned across jurisdictions according to verified economic deployment rather than only server location.

This concept would require coordination with existing international corporate tax rules and is proposed as a research mechanism rather than a claim of current law.

XI. ILLUSTRATIVE EXAMPLE

Assume an urban logistics company previously employs 100 human dispatchers and drivers. It introduces AI dispatch agents and autonomous delivery robots.

A conventional tax system observes:

- lower payroll;
- fewer payroll contributions;
- higher firm profit;
- larger investment in automated capital.

A simple robot tax may charge each robot a fixed annual amount. This ignores whether a robot replaced a worker, increased safety, or created new demand.

CivicDividendOS instead records task-origin activity.

Suppose the Factor-Origin Ledger estimates:

$$\phi_H = 0.30, \quad \phi_A = 0.25, \quad \phi_R = 0.30, \quad \phi_D = 0.05, \quad \phi_K = 0.10.$$

The AI and robot share of attributable value is therefore:

$$VA^{A,R} = 0.55V. \quad (28)$$

The SAC rate is then reduced for verified safety improvements, creation of new maintenance jobs, and retraining, but increased if labour displacement and economic-rent concentration are high.

Revenue is not paid by the robots. It is paid by the firm that owns or beneficially uses the automation.

Part of the revenue immediately funds transition insurance. Another portion enters the PAWF. Investment returns then provide dividends to citizens and finance future human skills.

The long-term allocation therefore becomes:

$$\text{Human} \rightarrow \text{Wages} + \text{Lower Labour Tax} + \text{Dividend},$$

$$\text{Owner} \rightarrow \text{Profit} - \text{Corporate Tax} - \text{SAC} + \text{Investment Return},$$

$$\text{Public} \rightarrow \text{SAC} + \text{Public Equity Returns},$$

$$\text{Future Workers} \rightarrow \text{Human Capability Endowment}.$$

XII. RESEARCH QUESTIONS

- RQ1** Can task-level factor-origin accounting estimate the value attributable to humans, AI agents, robots, data, and conventional capital more accurately than robot counts, imputed robot wages, or capital expenditure?
- RQ2** Can a substitution–augmentation classifier distinguish automation that displaces taxable labour from automation that primarily complements human workers?
- RQ3** Does a dynamic Social Automation Contribution preserve government revenue with less distortion to innovation than a fixed robot tax?
- RQ4** Can combining automation contributions with a Human Earned-Income Shield improve employment and after-tax wage outcomes relative to raising labour or consumption taxes?
- RQ5** Does a Public Automation Wealth Fund create a more durable and equitable distribution of AI-era wealth than transfers financed entirely from annual tax revenue?
- RQ6** What allocation among citizen dividends, worker transition, human-capability investment, and public services maximizes long-run social welfare?
- RQ7** How should cross-border AI-agent activity be apportioned when ownership, compute infrastructure, users, and economic deployment occur in different jurisdictions?
- RQ8** How robust is CivicDividendOS to tax avoidance, model misclassification, strategic task decomposition, automation offshoring, and rapid changes in AI capability?

XIII. EXPECTED CONTRIBUTIONS

The proposed research aims to make the following contributions:

- 1) A *Factor-Origin Ledger* for attributing value creation in mixed human–AI–robot production.
- 2) An *Autonomous Economic Activity Passport* that identifies machine economic activity without granting machines fictional legal taxpayer status.
- 3) A *Substitution–Augmentation Classifier* linking the fiscal treatment of automation to its actual effect on human work.
- 4) An *Automation Contribution Base* derived from attributable value added rather than robot counts or fictional machine salaries.
- 5) A dynamic *Social Automation Contribution* with explicit credits for augmentation, training, new-task creation, and broad social benefit.
- 6) A *Human Earned-Income Shield* that recycles part of automation revenue into lower fiscal burdens on human work.
- 7) A *Public Automation Wealth Fund* combining redistribution with predistribution by building public ownership of productive AI-era capital.
- 8) A four-channel distribution waterfall comprising citizen dividends, worker-transition insurance, human-capability investment, and public services.

- 9) A *Deployment Nexus* for research into jurisdictional allocation of cloud-based autonomous economic activity.
- 10) A fiscal digital twin for testing incidence, innovation, inequality, employment, public revenue, and international leakage before policy deployment.

XIV. EVALUATION METHODOLOGY

The framework can be evaluated through a heterogeneous-agent macroeconomic and urban-production simulation.

The simulated economy should contain:

- households with different income, skills, wealth, and automation exposure;
- firms with heterogeneous productivity and market power;
- conventional capital;
- AI-agent capital;
- robotic capital;
- government tax and transfer accounts;
- a public automation wealth fund; and
- multiple domestic and cross-border sectors.

CivicDividendOS should be compared with:

- 1) existing labour/capital tax system;
- 2) fixed robot tax;
- 3) broad capital-income tax increase;
- 4) UBI funded through general taxation;
- 5) automation tax plus targeted retraining;
- 6) sovereign/public wealth fund without factor-origin taxation; and
- 7) proposed integrated CivicDividendOS.

Evaluation metrics should include:

- GDP and productivity;
- AI/robot adoption;
- employment and hours;
- wage share and capital share;
- household disposable income;
- Gini coefficients for income and wealth;
- poverty rate;
- government revenue volatility;
- payroll/social-insurance funding;
- investment and innovation;
- firm entry and exit;
- public-fund value;
- citizen dividend per capita;
- worker-transition duration;
- human-capability investment;
- tax avoidance/leakage; and
- intergenerational wealth distribution.

Sensitivity analysis should vary AI costs, robot costs, substitution elasticities, market concentration, international mobility of capital, tax compliance, and the speed of technological progress.

XV. POLICY INTERPRETATION

CivicDividendOS does not assume that robots or AI should always face additional tax. It instead adopts four principles.

First, *machines are measured but owners are taxed*. Second, *substitution and augmentation receive different fiscal treatment*. Third, *part of automation revenue is converted into broadly owned capital rather than immediately spent*. Fourth, *human work should not remain the easiest tax base simply because it is administratively familiar*.

This produces a fiscal architecture that can evolve as the economy moves from human-dominant production toward increasingly mixed or capital-dominant production.

XVI. CONCLUSION

The AI and robotics era creates a distribution problem before it creates a purely technical tax problem. If increasingly autonomous productive systems generate a rising share of economic value while wages remain the principal source of household income and payroll taxation remains central to public finance, productivity can increase while purchasing power, social-insurance finance, and wealth distribution become more fragile.

The proposed CivicDividendOS framework addresses this mismatch by separating three questions that are frequently conflated: who created economic value, who is legally responsible for tax, and who should ultimately share in the returns.

The Factor-Origin Ledger answers the first question. The Autonomous Economic Activity Passport links the second to the legally responsible owner or operator. The Social Automation Contribution, Human Earned-Income Shield, Public Automation Wealth Fund, and four-channel distribution waterfall address the third.

The proposed system therefore does not seek to make robots “tax citizens.” Instead, it creates a measurable and adaptive fiscal bridge between an economy in which humans earn most productive income and one in which a larger share of value may be generated by autonomous capital. Its long-term objective is to ensure that technological productivity can remain privately innovative while also becoming broadly investable, taxable, and socially shareable.

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